QCert: Verifying Sixteen Quantum Circuit Completeness Results in One Agda Framework

XIAONING BIAN, Tsinghua University, China
YUAN FENG, Tsinghua University, China
SARAH MENG LI, University of Amsterdam and QuSoft, The Netherlands
JOHN VAN DE WETERING, University of Amsterdam and QuSoft, The Netherlands

A set of circuit equations is complete if any two circuits denoting the same linear map can be derived from each other using these equations, and minimal if no equation can be derived from the others. Establishing these properties takes long case analyses spanning hundreds of pages, making the details hard to check by hand. To this end, we extend the previous formalization framework to QCert, and use it to formalize the main circuit completeness theorems to date. This includes sixteen results, fifteen on completeness and one on minimality, over finite gate sets such as Clifford, CNOT-dihedral, Clifford+T, Clifford+CS, real Clifford+CH, and Toffoli–Hadamard, on qubits and on qudits of every odd prime dimension. QCert divides the work between humans and LLM coding agents. We first built its framework by specifying the syntax, semantics, and group presentations. Then, each paper's main theorem can be expressed uniformly in this framework. Next, an agent took over and formalized the proofs in Agda, following the papers' arguments. Agda checked every proof, so a human need only check the relation table, the gate semantics, and the theorem statement for each paper. Overall, QCert confirms the main results of each work and identifies gaps in two of the existing proofs. Its ability to do verification at scale makes QCert well suited to verifying quantum circuit completeness in higher dimensions.

1 Introduction

An equational theory for a family of quantum circuits is a set of equations between circuits used as rewrite rules. By applying these rules to local subcircuits, we can transform a circuit into an equivalent one. Such a theory should be *sound* because every rewrite preserves the underlying linear map. It should also be *complete* so that any equivalence between circuits can be derived. Additionally, we want it to be *minimal*, with no redundant rules. Equational rewriting supports circuit optimization [18, 22, 23, 26], equivalence checking [27], and hardware mapping [25]. Completeness ensures that every circuit equivalence can be derived within the theory, making it a powerful tool for reasoning about quantum programs. For example, Hadamard-Π [15] provides a complete equational framework for reasoning about reversible quantum programs. Since 2014, completeness results have been established for a wide range of circuit families such as Clifford [5, 19, 21, 24], CNOT-dihedral [1, 9], two-qubit Clifford+T [7], three-qubit Clifford+CS [8], Toffoli–Hadamard [2] and real Clifford+CH [12] circuits, and with continuous parameters, all quantum circuits [14]. Completeness has also been obtained for the group of two-level matrices underlying several of these circuit families [6, 11, 15, 16, 20]. Minimal theories have been given for six of these circuit fragments [10] and for all quantum circuits [13].

Completeness proofs often involve long case distinctions that are difficult to check by hand. A common strategy is to choose a unique normal form for each linear map and prove

Contact: bian@dal.ca, yuan_feng@tsinghua.edu.cn, sarah.li@uva.nl, john@vdwetering.name

that the rules normalize every circuit. Equivalent circuits then have the same normal form and can be derived from one another. Proving normalization requires handling each gate and every shape of normal form, often over hundreds of pages [12, 15, 19]. Redundant rules can then be removed without affecting completeness.

Language models can now write formal proofs of this size; a proof of Fermat's Last Theorem in Lean, of 13 million lines, was written in 11 days [3]. What still needs a person is the statement: a formal proof proves what its statement says, and someone must check that the statement, and the definitions under it, mean what was intended [17]. Generated proofs are not written to be read, so that check must stay small. Our approach is a *scaffold*: code that people write and check once, which fixes the form of every theorem built on it, so the per-paper check is short and the rest left to Agda. Its first layer is an Agda library of circuits and their presentations [4].

This work grew out of our formalization of the completeness theorem for qudit Clifford circuits [5]. There we set up a framework by hand, and brought in LLM coding agents only at the last stage, to finish the proof. That experience convinced us that all completeness results for quantum circuits could, and should, be formalized within one framework. This paper does so: we built such a scaffold for completeness theorems, and on it formalized sixteen papers (Table 1). Our contributions are threefold.

- QCert: an earlier library of words, circuits and presentations [4], extended with matrix semantics over the rings of exact synthesis and a framework for the independence of rules, giving every theorem the same form (Section 2);
- machine-checked proofs of the sixteen theorems: about 1,500 modules and 420,000 lines of Agda with no postulates, checked in about 40 minutes (Section 3);
- a gap in the completeness proof of Hadamard-Π and its repair, a rule omitted from a Toffoli–Hadamard theory, and misprints in the other papers (Section 4);
- a method in which people build the framework and state the theorems, LLM coding agents write the proofs, and Agda checks them (Section 5).

2 One Form for Every Theorem

A relation on an alphabet X generates a congruence $\approx$ on the words over X, and a semantics is a monoid homomorphism $[\![-]\!]$ from words to a monoid M. It is sound if related words have equal images and complete if words with equal images are related; then the relation presents the image of $[\![-]\!]$, and if every element of M is an image, it presents M. For circuits, M is a group of matrices, and the image is the group that the gates generate.

The earlier library [4] fixes this vocabulary: a presentation theorem is a record, Γ IsSubPresentationOf G, holding an interpretation of words in a group G with proofs that it is a group homomorphism on words modulo $\approx$ and injective, and Γ IsPresentationOf G adds that it is onto. It also fixes the syntax of circuits: a gate set is a family indexed by arity, a gate sits on the bottom wires of a circuit and a shift moves a circuit up a wire, and any relation stated at the width it is drawn on is extended by structural rules (an equation holds one wire up, gates on disjoint wires commute, a scalar is the same on every wire), which are sound once for every gate set. What this work adds is the semantic side: the rings of exact synthesis, $\mathbb{Z}[\frac{1}{\sqrt{2}}, i]$, $\mathbb{Z}[\frac{1}{2}, i]$, $\mathbb{Z}[\frac{1}{\sqrt{2}}]$ and $\mathbb{Z}[\frac{1}{2}]$, with their unitary and

orthogonal groups as groups of Agda's standard library, and operators on bit vectors with the Kronecker calculus, so that the matrix of a circuit is defined once for every gate set.

Every development ends in one line of this form. The real stabilizer circuits of Makary et al. [21], with the relations R1–R16 of their Figure 1 lifted to n wires, conclude

```
presentation : {n : ℕ} → (n VRel,_===_) IsSubPresentationOf U (2 ^ n)
```

and the two-level matrices of Hadamard-Π, with the relations of its Figure 6,

```
presentation : {n : ℕ} → (_===_ {n}) IsPresentationOf (O n)
```

To trust such a line a reader checks four things: the relation, as transcribed from the paper; the group, which comes from the layer and is checked once for all papers; the interpretation of each gate as a matrix; and the one line that gives the record that interpretation. For the real stabilizer development these come to under 500 of its 12,010 lines. A single root module, the place to start reading, states all sixteen main theorems in this form.

3 Sixteen Formalizations

Table 1 lists the sources, what we proved, the size of each development and its typechecking time. Every theorem is proved without hypotheses. The two-level presentations hold in every dimension (Greylyn's for $n \leq 4$, as in the thesis), and the circuit theories for every number of wires, except where the paper fixes it.

The papers use few proof methods, and in Agda each becomes a route with a fixed shape. *Exact synthesis*: an algorithm writes every matrix as a word, and completeness follows by well-founded induction on a level from a Main Lemma that closes a square for every state and generator; the induction is a module written once, and the author supplies the Main Lemma. *Normal forms*: every circuit is pushed to a normal form, and the normal form is read back from the matrix. *Induction on the width*: a circuit is decomposed into a part one wire down and a part on the bottom wire. *Encoding*: circuits are translated into a theory already known to be complete and back. And two ports reuse Bian and Selinger's own Agda proofs [7, 8], which relate the circuit relations to two-level relations by the Reidemeister–Schreier method, and connect them to our two-level presentations, so that they end in a statement about matrices. The qudit Clifford theorem of Bian et al. [5] is stated against an algebraic model of the Clifford group, built from the Pauli and symplectic groups with a phase; the others are stated against matrices.

Much of each development is generated. The derivations that a paper gives as computations, such as Selinger's 105-page supplement or the box relations of the qutrit normal form, come from scripts that follow the paper or search for the steps, and are checked by a rewriter in Agda; soundness is decided by evaluating matrices. What a paper gives as an idea, the Main Lemma or the uniqueness of a normal form, is written. In the real stabilizer development, half the lines are generated and a fifth are the paper's mathematics.

4 What We Found

A gap in Hadamard-Π. Hadamard-Π proves that the relations of its Figure 6 present the group $O_n(\mathbb{Z}[\frac{1}{\sqrt{2}}])$ of orthogonal matrices over $\mathbb{Z}[\frac{1}{\sqrt{2}}]$ (its Theorem 4.14), by the exact synthesis route. One subcase of its Main Lemma relies on Lemmas A.16 and A.17, and both

Source	Object	Result	Mod.	Time
Groups of two-level matrices				
Greylyn [2014]	$U_n(\mathbb{Z}[\frac{1}{\sqrt{2}}, i])$	presentation, $n \leq 4$	31	42
Bian and Selinger [2021]	$U_n(\mathbb{Z}[\frac{1}{2}, i])$	presentation	37	62
Li et al. [2021]	$O_n(\mathbb{Z}[\frac{1}{2}])$, L_n	presentations	68	79
Fang et al. [2026]	$O_n(\mathbb{Z}[\frac{1}{\sqrt{2}}])$	presentation	44	44
Clément [2019]	$O_n(\mathbb{Z}[\frac{1}{\sqrt{2}}])$	presentation	40	64
Circuits				
Selinger [2015]	Clifford	sub-presentation	31	34
Bian et al. [2026]	Clifford, odd prime p	presentation of a model	215	546
Makary et al. [2021]	real stabilizer	sub-presentation	30	35
Amy et al. [2018]	CNOT-dihedral	sub-presentation	30	24
Blake [2026a]	phase-affine, prime d	sub-presentations	56	61
Bian and Selinger [2023a]	two-qubit Clifford+T	sub-presentation in U_4	62	118
Bian and Selinger [2023b]	three-qubit Clifford+CS	sub-presentation in U_8	123	250
Clément [2026]	real Clifford+CH	sub-presentation	346	405
Amy et al. [2024]	three-qubit Toffoli–Hadamard	presentation of L_8	201	102
Li et al. [2025]	qutrit Clifford	presentation	89	282
Minimality				
Blake [2026b]	six fragments	minimal, complete	73	183
All			1,476	2,332

Table 1. The object presented, the result (a presentation of the whole group or of the group the gates generate), the number of modules, and typechecking time from scratch in seconds, the sum of the modules' own times given the developments they build on (Agda 2.8, Core i5-14400).

are false: each fails on a unit column of dimension 6, such as $(-1-\sqrt{2}, 1, -1-\sqrt{2}, 1, \sqrt{2}, -2+\sqrt{2})/\sqrt{2}^4$, because the proof drops a +1 from a bracket that it reduces modulo 2. Following the case analysis on random states, we found a state in dimension 9 at which every diagram the proof offers rises above the level it must stay below, so the proof of the Main Lemma does not cover it. The theorem is still true. An unpublished thesis of Clément from 2019 [11] presents the same group by other relations, with a case that covers exactly this edge; we proved the two relation sets equivalent in every dimension, formalized his case as a decision tree that Agda checks by evaluation, and then his whole proof. Clément's 2026 theorem on real Clifford+CH circuits [12] imports Theorem 4.14, so it rested on the same gap; our formalization of it discharges the import.

An omitted relation. Amy, Ross and Wesley [2] present three-qubit Toffoli–Hadamard circuits in two steps, first $O_8(\mathbb{Z}[\frac{1}{2}])$ by the Toffoli and $\mathrm{K} = \mathrm{H} \otimes \mathrm{H}$ gates, then the scaled dyadic orthogonal group L_8. Their Section 5.5 uses a relation r_{K} that defines $\mathrm{K}_{1,2}$ by a controlled K, but the figures that define the relations of their Theorem 5.2 omit it, and it does not follow from them: reading each Toffoli-K gate g as RgR, with $R = I - J/4$ for the

all-ones matrix J, satisfies every other relation and not r_{K}. So the relations as printed are not complete; with r_{K} restored, we prove that they present L_8.

Misprints. The other papers are correct in what we formalized, up to misprints, several of them in statements that proofs use: eleven of the 137 typed relations of Makary et al. [21], one relation of the L_n table of Li et al. [20], the scalar of a qutrit rule in Blake [10], two equations of the encoding of Clément [12], five formulas of Amy et al. [2], and one box relation of Li et al. [19]. Each was found because the statement failed a numerical check or a derivation failed at the step that used it.

5 How the Proofs Were Produced

Besides the qudit Clifford development, LLM coding agents wrote the other fifteen under our direction, within four weeks, most of them in less than a day each. Three habits made this possible. Every claim is checked numerically before it is formalized, with exact matrices in a short script, which caught most misprints before any Agda was written. Long derivations are generated, by scripts that follow the paper's own steps, so that a failing step points at a line of the paper. And the scaffold leaves the agent no choice about the statement: the relation, the group and the record are fixed, and only the proof remains.

The main cost was Agda's, not the agent's. The conversion checker compares terms by unfolding them, so two readings of a long product of matrices that are not syntactically equal can take exponential time; readings of circuits are therefore abstract, and equations between matrices are decided as one boolean computed by row operations, a hundred to four hundred times faster than comparing entry by entry. Everything is constructive, with no excluded middle, choice or real numbers, so every case split is a decision procedure and every proof is a program: with soundness, completeness and decidable matrix equality, each theory's word problem is decided by running the proof.

6 Conclusion and Future Work

All sixteen developments share one framework, and this matters in three ways. First, every theory, whatever its gate set or proof method, is stated with the same circuits, presentation records and matrix groups over the same rings: a common language in which rewrite rules for different circuit families can be related, as when we proved Clément's relations and Hadamard-Π's equivalent. Second, it lays the foundation for completeness in higher-dimensional systems, whose case analyses outgrow what a reader can check (the qutrit theorem rests on hundreds of box relations): soundness is decided by evaluating exact matrices, derivations are generated and machine-checked, and the corpus typechecks in about 40 minutes. Third, it paves the way for number-theoretic exact synthesis: its rings, their unitary and orthogonal groups and the route from a Main Lemma to completeness (Section 3) are formalized once, so a new synthesis algorithm can be verified efficiently, including its Main Lemma, where the gap in Hadamard-Π lay (Section 4).

Future work. We are porting the developments to Lean, which has a much larger community and whose library Mathlib covers far more of mathematics, so that our theorems

can be used alongside it. Going forward, we will use this framework to solve new problems in circuit completeness and exact synthesis, such as the Clifford completeness in all dimensions and optimising T-count inmulti-qubit Clifford+T exact synthesis.

References

[1] Matthew Amy, Jianxin Chen, and Neil J. Ross. 2018. A Finite Presentation of CNOT-Dihedral Operators. In *Proceedings of the 14th International Conference on Quantum Physics and Logic (QPL 2017) (Electronic Proceedings in Theoretical Computer Science, Vol. 266)*. 84–97.

[2] Matthew Amy, Neil J. Ross, and Scott Wesley. 2024. A Sound and Complete Equational Theory for 3-Qubit Toffoli-Hadamard Circuits. In *Proceedings of the 21st International Conference on Quantum Physics and Logic (QPL 2024) (Electronic Proceedings in Theoretical Computer Science, Vol. 406)*. 1–43. arXiv:2407.11152.

[3] Anthropic. 2026. Formalizing Fermat's Last Theorem. https://www.anthropic.com/research/formalizing-fermats-last-theorem. 4 September 2026. Lean code at https://github.com/anthropics/fermats-last-theorem.

[4] Xiaoning Bian and Yuan Feng. 2027. Normal Forms and Complete Relations for Circuits in Agda. *Proceedings of the ACM on Programming Languages* POPL (2027). To appear.

[5] Xiaoning Bian, Sarah Meng Li, Neil J. Ross, John van de Wetering, and Yuming Zhao. 2026. A Complete and Natural Rule Set for Multi-Qudit Clifford Circuits in All Odd Prime Dimensions. arXiv:2609.40106.

[6] Xiaoning Bian and Peter Selinger. 2021. Generators and Relations for $U_n(\mathbb{Z}[1/2, i])$. In *Proceedings of the 18th International Conference on Quantum Physics and Logic (QPL 2021) (Electronic Proceedings in Theoretical Computer Science, Vol. 343)*. 145–164. arXiv:2105.14047.

[7] Xiaoning Bian and Peter Selinger. 2023. Generators and Relations for 2-Qubit Clifford+T Operators. In *Proceedings of the 19th International Conference on Quantum Physics and Logic (QPL 2022) (Electronic Proceedings in Theoretical Computer Science, Vol. 394)*. 13–28. arXiv:2204.02217.

[8] Xiaoning Bian and Peter Selinger. 2023. Generators and Relations for 3-Qubit Clifford+CS Operators. In *Proceedings of the 20th International Conference on Quantum Physics and Logic (QPL 2023) (Electronic Proceedings in Theoretical Computer Science, Vol. 384)*. 114–126. arXiv:2306.08530.

[9] Colin Blake. 2026. Completeness for Prime-Dimensional Phase-Affine Circuits. arXiv:2603.06466.

[10] Colin Blake. 2026. Simpler Presentations for Many Fragments of Quantum Circuits. In *11th International Conference on Formal Structures for Computation and Deduction (FSCD 2026) (LIPIcs)*. To appear. arXiv:2602.09874.

[11] Alexandre Clément. 2019. A complete set of relations for the real Clifford+T group. Unfinished manuscript, March 2019, supervised by N. J. Ross and P. Selinger.

[12] Alexandre Clément. 2026. A Complete Equational Theory for Real-Clifford+CH Quantum Circuits. arXiv:2602.06644. Main text 13 pages, appendix of 264 pages of equational derivations.

[13] Alexandre Clément, Noé Delorme, and Simon Perdrix. 2024. Minimal Equational Theories for Quantum Circuits. In *Proceedings of the 39th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS 2024)*. arXiv:2311.07476.

[14] Alexandre Clément, Nicolas Heurtel, Shane Mansfield, Simon Perdrix, and Benoît Valiron. 2023. A Complete Equational Theory for Quantum Circuits. In *Proceedings of the 38th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS 2023)*. arXiv:2206.10577.

[15] Wang Fang, Chris Heunen, and Robin Kaarsgaard. 2026. Hadamard-Π: Equational Quantum Programming. *Proceedings of the ACM on Programming Languages* 10, POPL (2026), 5:1–5:31. arXiv:2506.06835.

[16] Seth E. M. Greylyn. 2014. Generators and relations for the group $U_4(\mathbb{Z}[1/\sqrt{2}, i])$. MSc thesis, Dalhousie University. arXiv:1408.6204.

[17] Thomas Hales. 2026. What Mathematicians Should Know about the Lean Theorem Prover: Questions of Reliability and AI. Guest post on Terence Tao's blog *What's new*, https://terrytao.wordpress.com/2026/10/09/what-mathematicians-should-know-about-the-lean-theorem-proverquestions-of-reliability-and-ai/. 9 October 2026.

[18] Kesha Hietala, Robert Rand, Shih-Han Hung, Xiaodi Wu, and Michael Hicks. 2021. A Verified Optimizer for Quantum Circuits. *Proceedings of the ACM on Programming Languages* 5, POPL (2021), 37:1–37:29.

[19] Sarah Meng Li, Michele Mosca, Neil J. Ross, John van de Wetering, and Yuming Zhao. 2025. A Complete and Natural Rule Set for Multi-Qutrit Clifford Circuits. In *Proceedings of the 22nd International Conference on Quantum Physics and Logic (QPL 2025) (Electronic Proceedings in Theoretical Computer Science, Vol. 426)*. 23–78. arXiv:2508.14670.

[20] Sarah Meng Li, Neil J. Ross, and Peter Selinger. 2021. Generators and Relations for the Group $O_n(\mathbb{Z}[1/2])$. In *Proceedings of the 18th International Conference on Quantum Physics and Logic (QPL 2021) (Electronic Proceedings in Theoretical Computer Science, Vol. 343)*. 210–264. arXiv:2106.01175.

[21] Justin Makary, Neil J. Ross, and Peter Selinger. 2021. Generators and Relations for Real Stabilizer Operators. In *Proceedings of the 18th International Conference on Quantum Physics and Logic (QPL 2021) (Electronic Proceedings in Theoretical Computer Science, Vol. 343)*. 14–36. doi:10.4204/EPTCS.343.2 arXiv:2109.05655. Ancillary supplement of 76 pages.

[22] Dmitri Maslov, Gerhard W. Dueck, D. Michael Miller, and Camille Negrevergne. 2008. Quantum Circuit Simplification and Level Compaction. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* 27, 3 (2008), 436–444. doi:10.1109/TCAD.2007.911334

[23] Yunseong Nam, Neil J. Ross, Yuan Su, Andrew M. Childs, and Dmitri Maslov. 2018. Automated optimization of large quantum circuits with continuous parameters. *npj Quantum Information* 4 (2018), 23. doi:10.1038/s41534-018-0072-4

[24] Peter Selinger. 2015. Generators and relations for n-qubit Clifford operators. *Logical Methods in Computer Science* 11, 2:10 (2015). arXiv:1310.6813.

[25] Seyon Sivarajah, Silas Dilkes, Alexander Cowtan, Will Simmons, Alec Edgington, and Ross Duncan. 2020. t|ket⟩: A Retargetable Compiler for NISQ Devices. *Quantum Science and Technology* 6, 1 (2020), 014003. doi:10.1088/2058-9565/ab8e92

[26] Mingkuan Xu, Zikun Li, Oded Padon, Sina Lin, Jessica Pointing, Auguste Hirth, Henry Ma, Jens Palsberg, Alex Aiken, Umut A. Acar, and Zhihao Jia. 2022. Quartz: Superoptimization of Quantum Circuits. In *Proceedings of the 43rd ACM SIGPLAN International Conference on Programming Language Design and Implementation (PLDI 2022)*. Association for Computing Machinery, New York, NY, USA, 625–640. doi:10.1145/3519939.3523433

[27] Shigeru Yamashita and Igor L. Markov. 2010. Fast Equivalence-checking for Quantum Circuits. *Quantum Information and Computation* 10, 9–10 (2010), 721–734. doi:10.26421/QIC10.9-10-1